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REVIEW

Umbrella review of low-level light therapy: evidence across clinical conditions

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Abstract

Low-level light therapy (LLLT) and photobiomodulation (PBM) have been studied in diverse clinical fields. Although numerous systematic reviews have been published, their findings are fragmented and inconsistent. This umbrella review of systematic reviews aimed to synthesize the evidence on the effectiveness and safety of LLLT across various clinical conditions. PubMed, Embase, Cochrane Library, and Web of Science databases were searched from inception to February 24, 2026. Eligible studies were systematic reviews or meta-analyses of LLLT in humans. Data on clinical conditions, intervention characteristics, overall direction of conclusions, and adverse events were extracted. Methodological quality was assessed using AMSTAR 2. A total of 100 systematic reviews were included, representing 1,164 primary-study occurrences and 790 unique eligible primary studies. Oral and dental conditions accounted for the largest evidence base (53 reviews), followed by musculoskeletal disorders (22 reviews), oncology-related complications (8 reviews), and wound and lymphatic conditions (8 reviews). Overall, 62 reviews reached conclusions favoring the evaluated intervention, 34 reported neutral or mixed conclusions, and 4 had unclear conclusions. The highest proportion of favorable conclusions was observed for oncology-related complications. Of the 100 reviews, 11 reported one or more adverse events or side effects, 32 explicitly reported no adverse events, and 57 did not provide adverse-event information. Most reviews received low (30.0%) or critically low (60.0%) overall confidence ratings using AMSTAR 2. The review literature suggests potential clinical benefit of LLLT in selected indications, particularly oncology-related complications such as oral mucositis. However, these descriptive patterns do not establish treatment efficacy or the certainty of the underlying evidence. Heterogeneity in treatment parameters, generally low methodological confidence, and incomplete adverse-event reporting limit interpretation and preclude firm conclusions about safety.

Keywords Light therapy · Photobiomodulation · Low–level light therapy · Umbrella review · Systematic review

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Introduction

Light-based therapies, including low–level light therapy (LLLT), photobiomodulation (PBM), and light-emitting diodes (LEDs), have attracted increasing attention as noninvasive approaches to modulate biological processes [1]. Light therapy is a light–dependent treatment modality for various diseases. By delivering specific wavelengths and energy doses, light therapy is proposed to modulate cellular metabolism, enhance tissue repair, reduce inflammation, and influence neural and immune functions [2]. These mechanisms underlie its wide range of clinical applications, spanning dentistry, dermatology, oncology, musculoskeletal medicine, and neurological or psychiatric disorders [3–7].

In particular, LLLT, which is typically delivered in the red to near–infrared spectrum, is widely used in medical practice for treating various diseases. LLLT uses low–power light from lasers or LEDs, typically at wavelengths of 600 to 1,100 nm in the red to near–infrared spectrum [8]. These wavelengths are intended to elicit photobiomodulatory responses and promote positive therapeutic responses [1, 9]. Previous studies indicate that LLLT is implemented to target tissues using specific wavelengths and optimized fluence (energy density) parameters to achieve therapeutic outcomes without inducing adverse effects [10]. The energy levels applied are sufficiently low to avoid thermal effects [7]. The noninvasive nature of LLLT has contributed to its clinical appeal, although its safety profile remains incompletely characterized because adverse events have not been consistently assessed or reported. Consequently, clinical interest has expanded globally, and numerous trials and systematic reviews have been published [11, 12].

However, despite the growing body of evidence, findings remain heterogeneous and occasionally contradictory. While some reviews report robust benefits of LLLT, others find equivocal or neutral effects [1, 10]. The diversity of light therapy parameters, including wavelength, energy density, treatment duration, and frequency, further complicates the interpretation and generalizability of the results [11].

Given this fragmented landscape, an umbrella review of systematic reviews may provide an opportunity to evaluate the breadth and credibility of the evidence [12]. This method allows for synthesis across diverse health conditions, assessment of the methodological rigor of existing systematic reviews, and identification of gaps in the literature on LLLT.

The objective of this umbrella review was to synthesize evidence from systematic reviews and meta-analyses on LLLT across clinical conditions, with a focus on effectiveness, adverse events, and methodological quality of the included reviews.

Methods

Protocol and reporting

This umbrella review followed the PRIOR statement for reporting overviews of reviews [13] and PRISMA statement for reporting systematic reviews [14]. The protocol was developed prospectively, with eligibility criteria, search methods, and data extraction procedures defined in advance, and was registered in the International Prospective Register of Systematic Reviews (PROSPERO: CRD420261327092).

Eligibility criteria

The following PICO framework and eligibility criteria were applied for this review:

Population: Human participants of any age, sex, or clinical condition were included.

Intervention: Non-thermal, low-intensity light-based therapies intended to induce photobiomodulatory effects, including low-level light therapy (LLLT), photobiomodulation (PBM), and light-emitting diode (LED) therapy, were eligible. Eligibility was determined based on intervention classification and therapeutic intent reported in the original reviews rather than predefined thresholds for wavelength, power output, or fluence because treatment parameters varied across indications and reporting was inconsistent. High-intensity light therapy, photodynamic therapy, cosmetic laser procedures, and diagnostic applications were excluded. In reviews containing both eligible and potentially ineligible interventions, eligibility was reassessed at the primary study level. Separable primary studies using thermal or ablative laser applications, including CO_2, Er: YAG, Er, Cr: YSGG, or high-energy Nd: YAG lasers, were excluded unless their use as non-thermal photobiomodulation could be clearly established.

Comparator: Eligible comparators included sham, placebo, standard care, and active interventions.

Outcomes: Primary outcomes included pain reduction, healing or disease severity, functional improvement, and structural or biological outcomes. Because meta-analytic effect estimates were not comparable across reviews, no umbrella-level quantitative effect estimate was calculated. Secondary outcomes included adverse events, side effects, and other harms associated with the intervention. Whether adverse events were reported and, when available, their types were summarized descriptively.

Inclusion criteria: Published systematic reviews (SRs) or meta–analyses of randomized or non–randomized controlled trials of light therapy in humans.

Exclusion criteria: Narrative or scoping reviews, non–systematic evidence syntheses, preclinical–only studies, or

articles not available in English. Reviews that did not use a Cochrane risk-of-bias tool to assess the included primary studies were also excluded.

Search strategy

A systematic search was conducted in PubMed, Embase, Cochrane Library, and Web of Science databases from inception to February 24, 2026. The strategy combined database-specific subject headings and free-text terms related to phototherapy, light therapy, LLLT, and photobiomodulation, with terms for systematic reviews and meta-analyses applied where appropriate. The complete database-specific search strategies are provided in Supplementary Material 1.

Reference lists of included reviews and related citations were manually searched for additional eligible studies.

Study selection

Two reviewers independently screened the titles, abstracts, and full texts according to the eligibility criteria for study selection. Disagreements were resolved by discussion or consultation with a third reviewer if necessary.

Data extraction

Two reviewers independently extracted the data using a standardized data extraction form. The following review–level information was extracted and tabulated:

- Bibliographic details (first author, year, country) of the review.
- Clinical conditions and categories (oral and dental conditions, musculoskeletal disorders, oncology-related complications, wound and lymphatic conditions, ocular conditions, dermatologic and hair conditions, respiratory conditions, auditory conditions).
- Number of included primary studies and reported sample size.
- Type of intervention (device type).
- Treatment characteristics (wavelength range, energy per point, fluence range, emission mode).
- Overall direction of review conclusions (favorable, neutral/mixed, or unclear).
- Adverse events or side effects related to the intervention.
- Primary outcome(s) reported in each included review.
- Identifiers of included primary studies for construction of the citation matrix and overlap assessment.

Where an included systematic review contained both eligible and ineligible primary studies, eligibility and extracted data were reassessed at the primary-study level. Only eligible primary studies contributed to the evidence counts, intervention characteristics, reported sample size, and overlap assessment. Reported sample size was extracted for eligible primary studies where separable and was summed descriptively across reviews. Because some reviews reported teeth or other units of analysis and some primary studies appeared in multiple reviews, the cumulative reported sample size was not interpreted as the number of unique participants.

Adverse-event reporting was classified into three mutually exclusive categories: one or more adverse events or side effects reported, no adverse events explicitly reported, or adverse-event information not reported. The absence of reported information was not interpreted as evidence that no adverse events occurred.

Quality assessment

The methodological quality of included reviews was appraised using AMSTAR 2 (A Measurement Tool to Assess Systematic Reviews) [15], which evaluates 16 items including protocol registration, search comprehensiveness, risk of bias assessment, and synthesis methods. Reviews were categorized as having high, moderate, low, or critically low confidence in their results, following the AMSTAR 2 guidance.

Data synthesis

The conclusions were synthesized narratively according to clinical categories. When multiple reviews addressed the same condition, their conclusions were descriptively compared. To descriptively map the direction of conclusions across the included reviews, we categorized the review authors' overall conclusions regarding treatment effects as favorable, neutral, or unclear. A favorable classification indicated that the review authors concluded that the intervention was beneficial; neutral indicated that the authors reported no clear benefit or mixed conclusions; and unclear indicated that no clear overall conclusion could be determined. This classification was based on the review authors' stated interpretations and was used solely to characterize the conclusions reported in the review literature. It was not considered a quantitative synthesis of efficacy, effect magnitude, or certainty of evidence.

Overlap assessment

The overlap assessment was performed to inform interpretation of the overall conclusions by identifying the extent to which primary studies were repeatedly included across systematic reviews. Primary study identifiers extracted from each systematic review were standardized using canonical study identifiers to account for differences in citation

formats and publication years, and a citation matrix was constructed. The corrected covered area (CCA) was then calculated to quantify the degree of overlap among primary studies included in multiple reviews.

Because an overall CCA may be diluted when reviews address unrelated clinical conditions, CCA was also calculated separately for each health condition category represented by more than one review. Overlap was not used as an exclusion criterion, and repeated inclusion of the same primary study was not interpreted as independent confirmation of treatment effects.

Pairwise CCA was also calculated for every pair of included reviews using the citation matrix. For two reviews, pairwise CCA equals the number of shared primary studies divided by the number of distinct primary studies in their combined set, multiplied by 100. Results were summarized separately for pairs within the same health condition category and pairs across categories.

Results

Overview of evidence base

A total of 100 reviews were included in this umbrella review (Fig. 1), representing 1,164 eligible primary-study occurrences and a cumulative reported sample size of 47,228 across reviews with reported values. After standardization and deduplication of study identifiers, 790 unique eligible primary studies were identified. The distribution of countries by the first authors showed that most reviews were from Brazil ($n=23$, 23.0%) and China ($n=21$, 21.0%), followed by India ($n=7$, 7.0%), Iran ($n=7$, 7.0%), and Chile ($n=6$, 6.0%). Smaller contributions came from the United States ($n=5$, 5.0%), United Kingdom ($n=4$, 4.0%) and Spain ($n=4$, 4.0%), with the remaining reviews distributed across several other countries. The reviews were published between 2006

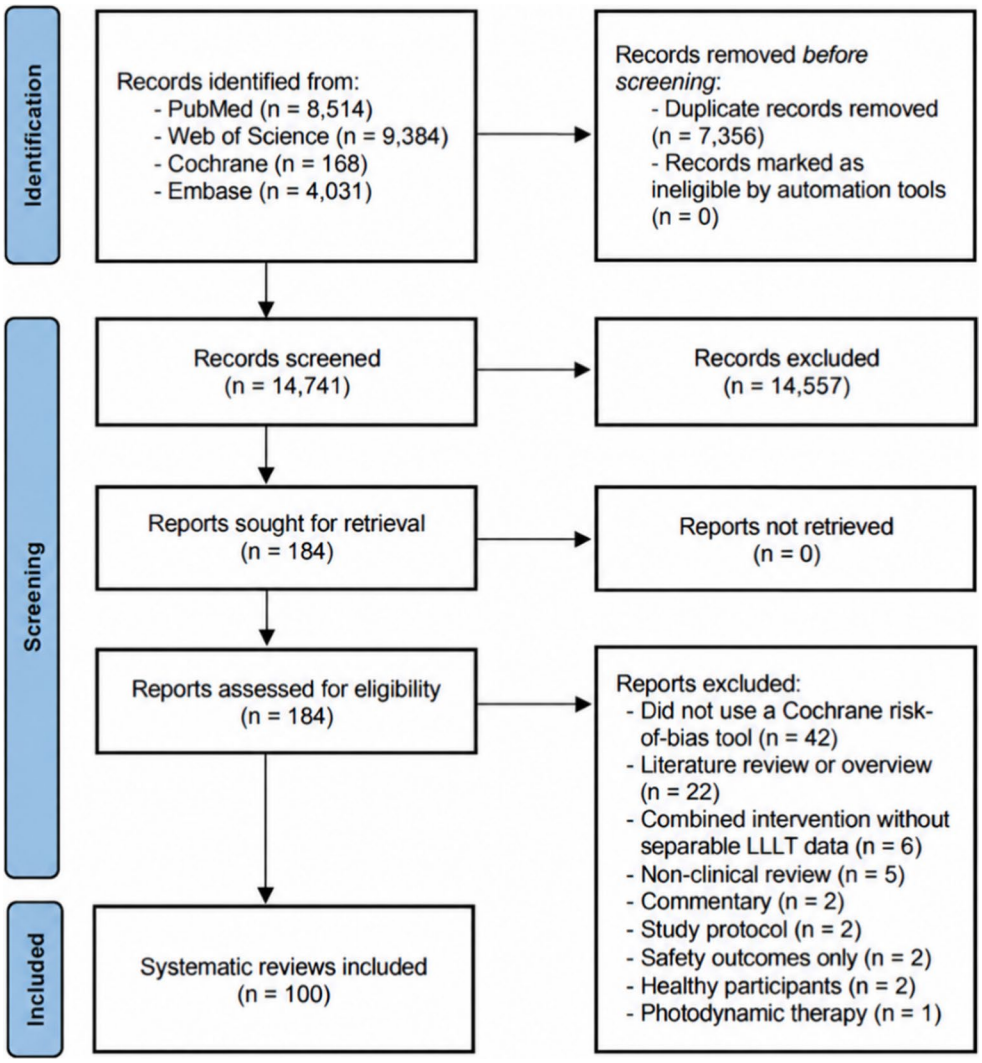


Fig. 1 Flow diagram of systematic review identification, screening, eligibility assessment, and inclusion

and 2026, with a median publication year of 2021 (IQR, 2019–2023). The most productive years were 2021, 2023, and 2025 ($n = 13$ each), followed by 2022 ($n = 12$).

When categorized according to the underlying health condition, the largest category was oral and dental conditions ($n = 53$, 53.0%), followed by musculoskeletal disorders ($n = 22$, 22.0%). Other categories included oncology-related complications ($n = 8$, 8.0%), wound and lymphatic conditions ($n = 8$, 8.0%), ocular conditions ($n = 4$, 4.0%), dermatologic and hair conditions ($n = 3$, 3.0%), respiratory conditions ($n = 1$, 1.0%), and auditory conditions ($n = 1$, 1.0%). This distribution indicates that more than half of the available review evidence on LLLT has focused on oral and dental applications, whereas non-oral indications were less frequently represented (Table 1).

Regarding the direction of review authors' conclusions, 62 reviews (62.0%) reached overall conclusions favoring the evaluated intervention, 34 (34.0%) reported neutral or mixed conclusions, and 4 (4.0%) had unclear conclusions. The highest within-category proportions of favorable conclusions were observed for oncology-related complications (87.5%), wound and lymphatic conditions (75.0%), and ocular conditions (75.0%) (Table 2).

Oral and dental conditions

Oral and dental conditions contributed the largest body of evidence, with 53 reviews [16–68] including 624 eligible primary-study occurrences representing 445 unique primary studies. Of these, 32 reviews (60.4%) reached favorable conclusions, 20 (37.7%) reported neutral conclusions, and 1 review (1.9%) was classified as unclear. Favorable conclusions were most commonly reported for post-surgical healing and pain control, recurrent herpes labialis, and some orthodontic applications. Neutral conclusions were reported across several indications, including oral lichen planus, gingival recession, implant stability, and some postoperative dental outcomes.

Musculoskeletal disorders

Twenty-two reviews [69–90] included 359 eligible primary-study occurrences representing 257 unique primary studies on musculoskeletal disorders, including temporomandibular disorders, tendinopathy, osteoarthritis, carpal tunnel syndrome, low back pain, fibromyalgia, and related pain conditions. Twelve reviews (54.5%) reached favorable conclusions, 9 (40.9%) reported neutral conclusions, and 1 (4.5%) was classified as unclear. Favorable conclusions were most commonly reported for pain reduction and functional improvement in selected musculoskeletal conditions, whereas neutral conclusions were frequently reported in reviews with substantial variation in treatment parameters, comparator interventions, and outcome assessment.

Oncology–Related complications

Eight reviews [91–98] included 46 unique primary studies on oncology-related complications, most commonly oral mucositis and xerostomia associated with chemotherapy or radiotherapy. Seven reviews (87.5%) reached favorable conclusions, and 1 (12.5%) review reported neutral conclusions. Favorable conclusions were mainly reported for reducing the incidence, severity, or duration of oral mucositis.

Dermatologic and hair conditions

Three reviews [99–101] included 11 unique primary studies in dermatologic and hair conditions. Two reviews (66.7%) reached favorable conclusions and 1 (33.3%) reported neutral conclusions. The two favorable reviews evaluated hair density outcomes, whereas the review evaluating hair count and diameter was neutral.

Wound care and lymphatic disorders

Eight reviews [102–109] included 46 unique primary studies on wound and lymphatic conditions, including diabetic foot ulcers, pressure ulcers, skin wounds, post-mastectomy lymphedema, and nipple trauma and pain. Six reviews (75.0%) reached favorable conclusions and 2 (25.0%) reported neutral conclusions. Favorable conclusions were primarily related to wound healing, tissue repair, or edema reduction.

Ocular conditions

Four reviews [110–113] included 22 unique primary studies related to ocular conditions, including myopia progression, childhood myopia, and age-related macular degeneration. Three reviews (75.0%) reached favorable conclusions and 1 (25.0%) was classified as unclear. Favorable conclusions were reported mainly in myopia-related applications.

Respiratory conditions

Respiratory conditions were represented by 1 review [114], including 1 primary study on tuberculosis. This review was classified as unclear.

Table 1 General characteristics of the included systematic reviews (n = 100)

First author, year	Country	Category	Health condition	No. of included primary studies	Reported sample size	Direction of review authors' conclusions	Device type	Diode subtype	Wavelength range (nm)	Fluence range (J/cm^2)	Energy per point (J)	Mode (continuous or pulsed)	Adverse events reported
Abesi 2023	Iran	Oral and dental conditions	Post-gingivectomy wound healing and pain	6	140	Favorable	Laser (diode)	Unspecified	588–940	4	NR	Continuous	NR
Ahmad 2021	India	Musculoskeletal disorders	Temporomandibular joint disorders	37	1409	Neutral	Laser (diode, He-Ne, Nd:YAG)	AlGaInP, GaAlAs, GaAs,	632.8–1064	1.5–112.5	NR	NR	NR
Ahmed 2020	India	Oral and dental conditions	Recurrent aphthous ulcers	5	330	Favorable	Laser (diode, Nd:YAG)	AlGaInP, unspecified	658–1064	NR	NR	NR	No
Akram 2018a	Pakistan	Oral and dental conditions	Oral lichen planus	5	240	Neutral	Laser (diode)	AlGaInP	630–970	1.5–120	NR	NR	NR
Akram 2018b	Pakistan	Oral and dental conditions	Gingival recession defects	4	100	Neutral	Laser (diode)	GaAlAs	588–810	4–15	NR	NR	NR
Al-Maweri 2018	Saudi Arabia	Oral and dental conditions	Recurrent herpes labialis	6	492	Favorable	Laser (diode, He-Ne)	GaAlAs	632.5–870	2.04–48	NR	NR	No
Alosaimi 2025	Saudi Arabia	Dermatologic and hair conditions	Androgenetic alopecia	4	165	Neutral	Unspecified LLLT	-	NR	NR	NR	NR	NR
Anschau 2019	Brazil	Oncology-related complications	Cancer therapy–induced oral mucositis	5	317	Favorable	Unspecified LLLT	-	830–970	2–6.5	NR	NR	No
Bakdach 2020	Syria	Oral and dental conditions	Orthodontic tooth movement	25	570	Favorable	Mixed (diode, LED)	GaAlAs, GaAs	618–980	0.065–150	0.2–2.3	Mixed	NR
Barbosa 2022	Brazil	Oral and dental conditions	Post-orthognathic surgery pain, oedema, and neurosensory disorders	15	322	Favorable	Laser (diode)	AlGaInP, GaAlAs, GaAs	660–980	1.5–157.5	1.2–9	NR	NR
Barros 2022	Brazil	Oral and dental conditions	Herpes labialis	8	931	Favorable	Unspecified LLLT	-	632.8–1072	2.04–48	NR	NR	NR
Bekhet 2017	Egypt	Musculoskeletal disorders	Carpal tunnel syndrome	8	473	Neutral	Laser (diode & unspecified)	GaAlAs	780–880	6–11	2.7–11	Mixed	No
Bitencourt Reis 2022	Brazil	Oral and dental conditions	Orthodontic tooth movement	9	164	Neutral	Mixed (LED, diode)	GaAlAs	618–980	5–67.06	1	NR	NR
Bittencourt 2017	Brazil	Oral and dental conditions	Neurosensory disorders after orthognathic surgery	3	50	Favorable	Laser (diode)	GaAlAs	NR	5–70	NR	NR	NR
Bonacina 2026	Brazil	Oral and dental conditions	Postoperative endodontic pain	8	609	Favorable	Laser (diode)	AlGaInP, GaAlAs, unspecified	660–980	6–590	NR	NR	NR

Table 1 (continued)

First author, year	Country	Category	Health condition	No. of included primary studies	Reported sample size	Direction of review authors' conclusions	Device type	Diode subtype	Wavelength range (nm)	Fluence range (J/cm^2)	Energy per point (J)	Mode (continuous or pulsed)	Adverse events reported
Cabrera Martimbianco 2020	Brazil	Musculoskeletal disorders	Achilles tendinopathy	4	119	Neutral	Laser (diode & unspecified)	GaAlAs	810–904	NR	0.9–3.0	NR	No
Carneiro 2022	Brazil	Oral and dental conditions	Tooth sensitivity after dental bleaching	6	212	Favorable	Unspecified LLLT	-	660–810	10–60	NR	NR	No
Chen 2019	China	Oral and dental conditions	Postoperative pain after root canal treatment	7	446	Neutral	Laser (diode)	AlGaInP	800–970	70–90	1.08–15	Mixed	No
Chen 2020	Taiwan	Auditory conditions	Tinnitus	11	670	Neutral	Unspecified LLLT	-	650–830	NR	NR	NR	Yes
Cheung 2020	China (Hong Kong)	Musculoskeletal disorders	Carpal tunnel syndrome	6	418	Neutral	Laser (diode)	AlGaInP, GaAlAs, GaAs	660–904	1–10	0.072–3.6	NR	Yes
da Silva Mira 2023	Brazil	Oral and dental conditions	Pediatric dental conditions	4	257	Favorable	Unspecified LLLT	-	786.94–980	3–33.5	1–4	NR	NR
de Figueiredo Costa 2021	Brazil	Oral and dental conditions	Orthodontic mini-implant stability	6	108	Favorable	Laser (diode)	Unspecified	618–940	4–36	NR	NR	NR
de la Barra Ortiz 2025	Chile	Musculoskeletal disorders	De Quervain tenosynovitis	8	320	Favorable	Laser: diode, unspecified	GaAlAs	830–904	0.25–20	3–24	Continuous	NR
de la Barra Ortiz 2026	Chile	Musculoskeletal disorders	Musculoskeletal disorders	22	1352	Neutral	Laser: diode, unspecified	Unspecified	775–980	4–50	NR	Mixed	No
de Pedro 2020	Spain	Oral and dental conditions	Neuropathic orofacial pain	13	405	Favorable	Laser (diode, He-Ne, unspecified)	GaAlAs	630–980	2–105	NR	NR	No
Deana 2017	Chile	Oral and dental conditions	Orthodontic pain	20	788	Favorable	Laser (diode)	GaAlAs, GaAs	780–940	1–160	0.2–12	Mixed	No
Delaney 2018	USA	Dermatologic and hair conditions	Androgenetic alopecia	5	523	Favorable	Unspecified LLLT	-	630–660	47.9–67.3	NR	NR	Yes
Deng 2024	China	Ocular conditions	Myopia progression	6	793	Favorable	Unspecified LLLT	-	650	NR	NR	NR	Yes
Diaz 2025a	Chile	Oral and dental conditions	Maxillofacial neuropathies	18	1220	Favorable	Unspecified LLLT	-	630–980	2–20	NR	NR	Yes
Diaz 2025b	Chile	Musculoskeletal disorders	Temporomandibular disorders	44	1816	Favorable	Unspecified LLLT	–	632.8–1064	0.13–60	3–18	Mixed	Yes

Table 1 (continued)

First author, year	Country	Category	Health condition	No. of included primary studies	Reported sample size	Direction of review authors' conclusions	Device type	Diode subtype	Wavelength range (nm)	Fluence range (J/cm^2)	Energy per point (J)	Mode (continuous or pulsed)	Adverse events reported
Doiphode 2024	India	Oral and dental conditions	Primary molar pulpotomy	14	462	Neutral	Laser (diode & unspecified)	Unspecified	632–980	2.5–6.7	NR	Mixed	NR
Domah 2021	United Kingdom	Oral and dental conditions	Third molar surgery postoperative morbidity	17	1094	Neutral	Laser (diode, He-Ne)	GaAlAs, unspecified	550–940	3–12	6–106	Mixed	NR
Domingos de Barros 2022	Brazil	Oral and dental conditions	Postoperative pain after lower third molar extraction	15	648	Favorable	Laser (diode)	GaAlAs, unspecified	550–980	3–480	NR	NR	NR
dos Santos 2021	Brazil	Wound and lymphatic conditions	Diabetic foot ulcers	13	361	Favorable	Unspecified LLLT	-	400–904	1.5–10	NR	NR	No
Duarte de Oliveira 2021	Brazil	Oral and dental conditions	Third molar surgery postoperative pain, edema, trismus	10	308	Neutral	Laser (diode, He-Ne)	AlGaInP, unspecified	660–940	3–187.5	6–180	NR	NR
Fan 2024	China	Musculoskeletal disorders	Knee osteoarthritis	12	631	Favorable	Mixed (LED, diode)	NR	650–1,060	1.5–382	NR	NR	NR
Farzan 2021	Iran	Oral and dental conditions	Orthodontic separator placement pain	4	129	Favorable	Laser (diode)	GaAlAs	810–830	2–6	NR	NR	NR
Farzan 2022	Iran	Oral and dental conditions	Bone formation in rapid palatal expansion	4	85	Favorable	Laser (diode)	AlGaInP, GaAlAs, unspecified	660–980	140–238.85	3–6	NR	NR
Ferreira Gonzalez 2025	Spain	Oral and dental conditions	Third molar extraction complications	18	585	Neutral	Laser (diode)	Unspecified	660–940	3–212	4–50	NR	NR
Firoozi 2020	Iran	Oral and dental conditions	Neurosensory disturbance after sagittal split ramus osteotomy (SSRO)	8	188	Neutral	Mixed (LED, diode)	GaAlAs	632–980	1.5–100	1.2–9	NR	No
Franco 2023	Italy	Oncology-related complications	Oral mucositis after hematopoietic stem cell transplantation	3	98	Favorable	Unspecified LLLT	-	NR	NR	NR	NR	No
Gaitero 2025	Brazil	Wound and lymphatic conditions	Nipple trauma and pain	3	193	Neutral	Unspecified LLLT	–	660	5-66.6	0.2–3	NR	Yes
Gao 2025	China	Oral and dental conditions	Orthodontic pain	23	1007	Favorable	Unspecified LLLT	–	808–830	1–4	NR	NR	NR
Ge 2015	China	Oral and dental conditions	Orthodontic tooth movement acceleration	8	121	Neutral	Laser (diode, He-Ne)	GaAlAs, GaAs	632–860	2.5–25	NR	NR	No

Table 1 (continued)

First author, year	Country	Category	Health condition	No. of included primary studies	Reported sample size	Direction of review authors' conclusions	Device type	Diode subtype	Wavelength range (nm)	Fluence range (J/cm^2)	Energy per point (J)	Mode (continuous or pulsed)	Adverse events reported
Gentile 2024	Italy	Dermatologic and hair conditions	Androgenetic alopecia (male and female pattern hair loss)	7	607	Favorable	Mixed (LED, diode)	Unspecified	630–660	47.9–67.3	NR	NR	Yes
Ghaderi 2025	Iran	Oral and dental conditions	Dental pain	5	207	Favorable	Laser: diode	GaAlAs, Unspecified	660–980	4–69	2–16.25	Continuous	No
Gil 2025	Chile	Oral and dental conditions	Oral surgery	17	731	Favorable	Laser: diode, Nd: YAG, unspecified	GaAlAs, InGaAIP, Unspecified	660–1064	3–35	NR	Mixed	NR
Golez 2022	Slovenia	Oncology-related complications	Xerostomia/hyposalivation	18	558	Favorable	Mixed (LED, diode, He-Ne)	Unspecified	625–940	1.5–29.5	NR	Mixed	No
Grajales 2023	Spain	Oral and dental conditions	Orthodontic tooth movement acceleration	19	305	Favorable	Laser (diode)	GaAlAs, GaAs	658–980	4.2–25	NR	NR	NR
Guedes Pereira Máximo 2022	Brazil	Musculoskeletal disorders	Temporomandibular disorder	10	438	Favorable	Laser (diode & unspecified)	AlGaInP, GaAlAs, unspecified	630–904	1–105	6	NR	NR
He 2013	China	Oral and dental conditions	Orthodontic pain	6	402	Neutral	Laser (diode, He-Ne)	GaAlAs, unspecified	632.8–830	0.45–2.5	NR	Continuous	No
He 2015	China	Oral and dental conditions	Postoperative complications after mandibular third molar surgery	12	193	Favorable	Laser (diode, He-Ne, unspecified)	GaAlAs, unspecified	632.8–980	4–10	0.96–54	Mixed	No
He 2018	China	Oncology-related complications	Chemotherapy-induced oral mucositis	8	373	Favorable	Laser (diode & unspecified)	AlGaInAs, AlGaInP, GaAlAs	660–970	4–72	0.35–2.4	NR	NR
Henein 2021	United Kingdom	Ocular conditions	Age-related macular degeneration	2	132	Unclear	Mixed (LED, diode)	Unspecified	590–850	4–9	NR	Continuous	NR
Jajarm 2018	Iran	Oral and dental conditions	Oral lichen planus	7	NR	Neutral	Unspecified LLLT	-	NR	NR	NR	NR	NR
Jimenez 2024	USA	Musculoskeletal disorders	Spastic cerebral palsy	8	NR	Favorable	Unspecified LLLT	-	NR	NR	NR	Continuous	NR
Jing 2021	China	Musculoskeletal disorders	Temporomandibular joint disorders	16	538	Favorable	Laser (diode, He-Ne, Nd: YAG, unspecified)	AlGaInP, GaAlAs, GaAs	632.8–1064	NR	NR	NR	NR

Table 1 (continued)

First author, year	Country	Category	Health condition	No. of included primary studies	Reported sample size	Direction of review authors' conclusions	Device type	Diode subtype	Wavelength range (nm)	Fluence range (J/cm^2)	Energy per point (J)	Mode (continuous or pulsed)	Adverse events reported
Jnaneshwar 2023	India	Oral and dental conditions	Orthodontic tooth movement	10	223	Favorable	Laser (diode) & unspecified LLLT	GaAlAs, unspecified	618–980	2.25–150	NR	Continuous	NR
Joseph 2025	Finland	Oncology-related complications	Oral mucositis in cancer patients	5	256	Favorable	LED	InGaAIP, GaAIP, unspecified	600–670	0.99–4	0.24–2.34	Continuous	NR
Li 2015	China	Oral and dental conditions	Orthodontic pain	11	698	Neutral	Laser (diode, He-Ne)	AlGaInP, GaAlAs, GaAs, unspecified	632.8–910	0.5–6	0.042–10.88	NR	NR
Li 2018	China	Wound and lymphatic conditions	Diabetic foot ulcer	7	194	Favorable	Unspecified LLLT	-	400–904	2–10	NR	NR	No
Lourinho 2023	Brazil	Musculoskeletal disorders	Rheumatoid arthritis	18	793	Neutral	Laser (diode, He-Ne, unspecified)	AlAs, GaAlAs	632.5–1060	0.0075–25	1.55–7.5	NR	Yes
Lu 2023	China	Oral and dental conditions	Burning mouth syndrome	14	550	Neutral	Laser (diode, Nd: YAG, unspecified)	GaAlAs, unspecified	630–1064	1–200	2–6	Mixed	No
Lu 2025	China	Oral and dental conditions	Oral lichen planus pain	12	476	Favorable	Unspecified LLLT	–	630–980	0.3–177	NR	NR	No
Machado 2017	Brazil	Wound and lymphatic conditions	Pressure ulcers	4	210	Neutral	Laser (diode & unspecified)	GaAlAs, GaAs	650–950	1–4.5	NR	NR	NR
Malik 2023	India	Musculoskeletal disorders	Knee osteoarthritis	13	774	Favorable	Mixed (LED, diode)	GaAs, GaAlAs, unspecified	640–905	0.12–200	1.2–30	NR	NR
Meneses-Santos 2022	Brazil	Oral and dental conditions	Postoperative outcomes after orthognathic surgery	5	190	Neutral	Laser (diode)	InGaAsP, GaAlAs	660–940	3–100	1.2–2.8	Mixed	NR
Michelogiannakis 2022	USA	Oral and dental conditions	Orthodontic mini-screw implant stability	6	103	Favorable	Laser (diode)	Unspecified	618–940	NR	NR	Pulsed	NR
Mirzaei 2019	Germany	Oral and dental conditions	Inferior alveolar nerve sensory recovery after sagittal split osteotomy	7	181	Favorable	Laser (diode)	GaAlAs	660–930	1.5–100	1.2–9	NR	NR

Table 1 (continued)

First author, year	Country	Category	Health condition	No. of included primary studies	Reported sample size	Direction of review authors' conclusions	Device type	Diode subtype	Wavelength range (nm)	Fluence range (J/cm^2)	Energy per point (J)	Mode (continuous or pulsed)	Adverse events reported
Munguia 2018	USA	Musculoskeletal disorders	Temporomandibular myofascial pain	8	225	Neutral	Laser (diode, Nd:YAG)	AlGaInP, GaAlAs, GaAs	660–1064	3.4–105	NR	Mixed	NR
Nascimento Ribeiro 2024	Brazil	Oncology-related complications	Xerostomia in head and neck radiotherapy patients	4	126	Neutral	Laser (diode)	AlGaInP, GaAlAs	660–830	3–71	0.28–2	NR	NR
Peralta-Mamani 2019	Brazil	Oncology-related complications	Oral mucositis due to radiotherapy for head and neck cancer	7	363	Favorable	Laser (diode, He-Ne)	AlGaInP	632.8–685	1.3–6.3	0.05–3	NR	NR
Prados-Frutos 2016	Spain	Oral and dental conditions	Dental implant	2	20	Unclear	Laser (diode)	GaAlAs	637–830	6.26–92.1	0.25	Mixed	NR
Radithia 2024	India	Oral and dental conditions	Recurrent aphthous stomatitis	8	424	Favorable	Laser (diode, Nd:YAG)	AlGaInP, GaAlAs, unspecified	645–1064	2–12.9	0.1–3	Mixed	NR
Rankin 2017	United Kingdom	Musculoskeletal disorders	Carpal tunnel syndrome	22	1153	Neutral	Laser (diode, He-Ne, Nd:YAG)	GaAlAs, GaAs, unspecified	632.7–1064	3.4–11	1.2–8	Mixed	No
Reis Guerreiro 2021	Brazil	Oral and dental conditions	Postoperative endodontic pain	12	736	Favorable	Mixed (LED, diode)	AlGaIn, GaAlAs, unspecified	640–970	3–90	2.5	Mixed	NR
Ren 2015	China (Hong Kong)	Oral and dental conditions	Orthodontic pain	14	659	Neutral	Laser (diode)	AlGaInP, GaAlAs, GaAs, unspecified	635–910	0.5–80	0.18–9	Mixed	No
Ren 2022	China	Musculoskeletal disorders	Temporomandibular disorders pain	27	969	Favorable	Unspecified LLLT	-	632.8–1100	1.5–105	1.5–2	NR	NR
Shafaee 2023	Iran	Oral and dental conditions	Oral health in fixed orthodontic patients	6	142	Neutral	Laser (diode)	GaAlAs, unspecified	660–940	NR	2–5.7	Continuous	NR
Shan 2021	China	Oral and dental conditions	Dentin hypersensitivity	35	574	Favorable	Laser (diode, Nd:YAG) & unspecified LLLT	GaAlAs, unspecified	630–1067	2–100	0.1–300	Continuous	NR
Skondra 2018	Greece	Oral and dental conditions	Bone healing and regeneration after rapid maxillary expansion	4	93	Favorable	Laser (diode)	GaAlAs, unspecified	660–830	10–140	NR	NR	NR

Table 1 (continued)

First author, year	Country	Category	Health condition	No. of included primary studies	Reported sample size	Direction of review authors' conclusions	Device type	Diode subtype	Wavelength range (nm)	Fluence range (J/cm^2)	Energy per point (J)	Mode (continuous or pulsed)	Adverse events reported
Sobral de Oliveira-Souza 2023	Brazil	Oral and dental conditions	Orofacial pain	89	NR	Favorable	Mixed (LED, diode, He-Ne, Nd: YAG) & unspecified LLLT	AlGaInP, GaAlAs, GaAs, unspecified	632.8–1064	3–112.5	NR	NR	NR
Souza de Lima 2020	Brazil	Oncology-related complications	Oral mucositis prevention in head and neck cancer patients	4	500	Favorable	Laser (diode, He-Ne)	AlGaInP, GaAlAs	632.8–660	2.5–4	0.1–3.48	NR	NR
Stausholm 2019	Norway	Musculoskeletal disorders	Knee osteoarthritis	22	1063	Favorable	Unspecified LLLT	-	785–904	NR	0.054–8	NR	No
Taha 2024	United Kingdom	Wound and lymphatic conditions	Skin wounds	18	670	Favorable	Laser (diode) & unspecified LLLT	AlGaInP, GaAlAs, unspecified	400–890	1.04–13	NR	Mixed	NR
Tang 2023	China	Ocular conditions	Myopia	13	1857	Favorable	Unspecified LLLT	-	635–650	NR	NR	NR	No
Tchanque-Fossuo 2016	USA	Wound and lymphatic conditions	Diabetic foot ulcer	4	131	Favorable	Unspecified LLLT	-	400–890	2–43.2	NR	NR	No
Tripodi 2021	Australia	Musculoskeletal disorders	Tendinopathy	17	835	Neutral	Unspecified LLLT	-	810–905	0.5–5	NR	NR	No
Vande 2022	India	Oral and dental conditions	Dental implant stability	10	156	Neutral	Mixed (LED, diode, unspecified)	GaAlAs, unspecified	626–980	4–92.1	NR	NR	NR
Vlassov 2006	Russia	Respiratory conditions	Tuberculosis	1	130	Unclear	Laser (diode)	GaAs	890	NR	NR	NR	NR
Wahid 2024	Indonesia	Wound and lymphatic conditions	Post-mastectomy lymphedema	9	299	Favorable	Unspecified LLLT	-	NR	NR	NR	NR	No
Wang 2019	China	Musculoskeletal disorders	Plantar fasciitis	6	274	Favorable	Laser (diode)	GaAlAs, GaAs, unspecified	635–904	NR	5.6–16.8	NR	No
Yang 2025	China	Ocular conditions	Childhood myopia	8	1479	Favorable	Unspecified LLLT	-	650	NR	NR	NR	Yes
Yap 2025	Singapore	Musculoskeletal disorders	Pain in tendinopathy	35	1551	Favorable	Laser	-	633–1060	0.38–30	3–16	Mixed	NR

Table 1 (continued)

First author, year	Country	Category	Health condition	No. of included primary studies	Reported sample size	Direction of review authors' conclusions	Device type	Diode subtype	Wavelength range (nm)	Fluence range (J/cm^2)	Energy per point (J)	Mode (continuous or pulsed)	Adverse events reported
Yeh 2019	Taiwan	Musculoskeletal disorders	Fibromyalgia	9	325	Favorable	Mixed (LED, diode) & unspecified LLLT	GaAlAs, GaAs	640–950	2–4	2–4	Mixed	No
Yousefi-Nooraie 2008	Canada	Musculoskeletal disorders	Non-specific low back pain	7	384	Unclear	Laser (diode, Nd: YAG, unspecified)	GaAlAs, GaAs, unspecified	632–1060	NR	0.1–29.9	Mixed	No
Zhao 2021	China	Oral and dental conditions	Postoperative pain and wound healing after periodontal surgery	13	394	Neutral	Mixed (LED, diode)	GaAlAs, unspecified	588–940	4–60	NR	NR	Yes
Zheng 2023a	China	Oral and dental conditions	Orthodontic dental alignment	8	369	Favorable	Mixed (LED, diode)	GaAlAs, InGaAs	635–980	2.25–216	NR	Continuous	NR
Zheng 2023b	China	Oral and dental conditions	Orthodontic miniscrew stability	3	46	Neutral	Mixed (LED, diode)	Unspecified	618–940	36	NR	NR	NR
Zhou 2021	Singapore	Wound and lymphatic conditions	Diabetic foot ulcers	12	480	Favorable	Unspecified LLLT	-	400–904	0.18–10	NR	Mixed	No

Table 2 Evidence size and direction of review authors' conclusions by health condition category

Health condition category	No. of reviews	No. of primary study occurrences	No. of unique primary studies	Cumulative reported sample size	Direction of review authors' conclusions (review count, %)
Oral and dental conditions	53	624	445	19,883	Favorable: 32 (60.4%), Neutral: 20 (37.7%), Unclear: 1 (1.9%)
Musculoskeletal disorders	22	359	257	15,860	Favorable: 12 (54.5%), Neutral: 9 (40.9%), Unclear: 1 (4.5%)
Oncology-related complications	8	54	46	2,591	Favorable: 7 (87.5%), Neutral: 1 (12.5%)
Wound and lymphatic conditions	8	70	46	2,538	Favorable: 6 (75.0%), Neutral: 2 (25.0%)
Ocular conditions	4	29	22	4,261	Favorable: 3 (75.0%), Unclear: 1 (25.0%)
Dermatologic and hair conditions	3	16	11	1,295	Favorable: 2 (66.7%), Neutral: 1 (33.3%)
Respiratory conditions	1	1	1	130	Unclear: 1 (100.0%)
Auditory conditions	1	11	11	670	Neutral: 1 (100.0%)
Total	100	1,164	790*	47,228	Favorable: 62 (62.0%), Neutral: 34 (34.0%), Unclear: 4 (4.0%)

Percentages are calculated within each health condition category

Reported sample size is presented as reported for eligible primary studies and may refer to participants, teeth, or other units of analysis. Three reviews did not report a total sample size. Review-level sample sizes were summed across reviews and may include the same participants more than once because some primary studies were represented in multiple reviews. Accordingly, the cumulative reported sample size of 47,228 should not be interpreted as the number of unique participants. Category-level primary-study occurrences were summed across reviews, whereas unique primary studies were deduplicated separately within each health condition category. Direction categories reflect the conclusions reported by the review authors

* Category-specific unique primary-study counts are not mutually exclusive because some primary studies were represented in systematic reviews assigned to more than one category. After accounting for this cross-category overlap, the overall number of unique primary studies was 790

Auditory conditions

Auditory conditions were represented by 1 review [115], including 11 primary studies on tinnitus. This review was classified as neutral.

Primary outcome domains

The primary outcomes reported across the included systematic reviews were categorized into five outcome domains. Pain was the most frequently evaluated primary outcome domain (57 reviews), followed by functional outcomes (27 reviews), healing or disease severity (26 reviews), structural outcomes (19 reviews), and biological outcomes (3 reviews). The full list of primary outcomes reported by the included systematic reviews is provided in Supplementary Table S1.

Distribution of review conclusions across health conditions

A bubble map was constructed to visualize the distribution of review conclusions across health condition categories (Fig. 2). The x-axis represents the direction of review authors' conclusions (favorable, neutral, or unclear), and the y-axis lists the health condition categories, with the number of included reviews in parentheses. Bubble size reflects the number of reviews, while the labels show the review count and within-category percentage.

Adverse events

Fifty-seven reviews (57.0%) did not provide adverse-event information. Among the 43 reviews that addressed adverse events or side effects, 32 reviews (32.0% of all reviews) explicitly reported no adverse events, whereas 11 (11.0%) reported one or more events (Table 3). Reported events included local discomfort and skin reactions, such as erythema, warmth, burning, tingling, pruritus, dry skin, scalp irritation, acne, and pain. Other reported events included sudden deafness with dizziness, increased disease activity, and prolonged afterimage duration. However, the included reviews generally did not provide sufficient denominators, comparator-group data, causality assessments, standardized severity classifications, or absolute risks. Therefore, the frequency, comparative risk, severity, and relationship of these events to LLLT could not be determined.

Methodological quality

Overall methodological confidence in the included reviews was generally low according to AMSTAR 2 (Table 4). Sixty reviews (60.0%) were rated as critically low, 30 (30.0%) as low, 1 (1.0%) as moderate, and 9 (9.0%) as high confidence.

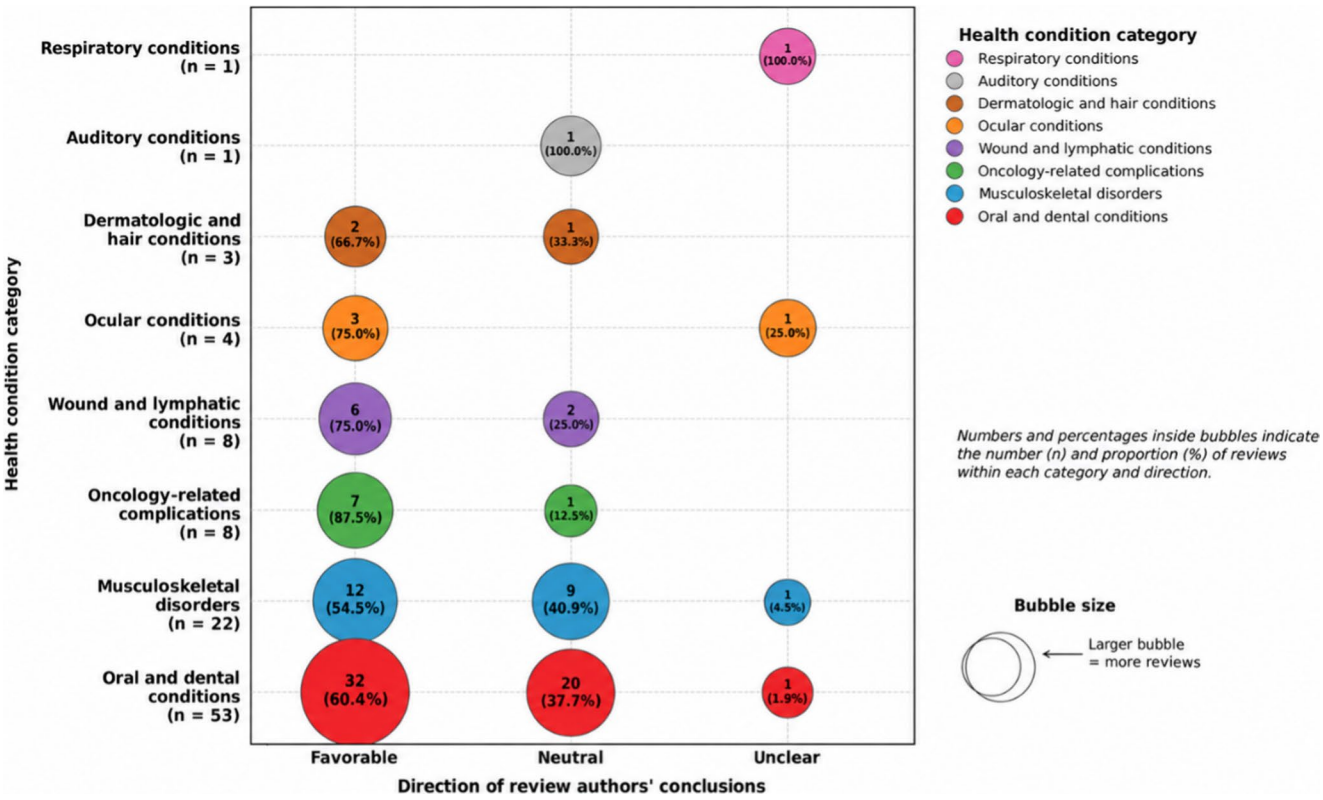


Fig. 2 Bubble map of the direction of review conclusions across health condition categories

Item-level performance varied considerably. Description of included studies (Q8) was rated Yes in 97 reviews and Partial Yes in 3, while PICO/inclusion criteria (Q1) was rated Yes in 99 reviews. Justification of eligible study designs (Q3) was rated Yes in 97 reviews. High adherence was also observed for disclosure of conflicts of interest (Q16, 93.0%), duplicate study selection (Q5, 90.0%), and duplicate data extraction (Q6, 86.0%). In contrast, reporting of funding for the included primary studies (Q10) was rated Yes in only 6.0% of reviews.

Among the AMSTAR 2 critical domains, performance was lowest for provision of an excluded-study list with justification (Q7; 22.0% Yes) and assessment of publication bias (Q15; 23.0% Yes or Partial Yes). Ten reviews met all applicable critical domains; 9 were rated as high confidence and 1 as moderate confidence. The item-level judgments and overall confidence ratings, together with graphical summaries of the AMSTAR 2 domain-level judgments and overall confidence, are presented in Supplementary Material 2.

Overlap assessment

The citation matrix comprised 790 unique eligible primary studies and 1,164 primary-study occurrences across the included systematic reviews. The overall CCA was 0.48%. However, category-specific CCA values were used for interpretation, since the overall CCA value combined reviews addressing diverse unrelated conditions. CCA was 22.73% for dermatologic and hair conditions, 10.61% for ocular conditions, 7.45% for wound and lymphatic conditions, 2.48% for oncology-related complications, 1.89% for musculoskeletal disorders, and 0.77% for oral and dental conditions. CCA was not applicable to the auditory and respiratory categories because each contained only one review. The citation matrix is provided in Supplementary Table S2. Among 1,674 within-category review pairs, 144 shared at least one primary study. Pairwise CCA ranged from 0% to 71.43%, indicating substantial overlap between some individual reviews despite the low overall CCA. Fifteen cross-category pairs also shared primary studies. Full pairwise results are provided in Supplementary Table S3.

Discussion

This umbrella review consolidated evidence from 100 systematic reviews on LLLT across diverse clinical conditions. A substantial proportion of the included reviews reached conclusions favoring the evaluated interventions, suggesting potential clinical benefit in selected indications. Nevertheless, the direction of review-level conclusions alone does not establish treatment efficacy or the certainty of the

Table 3 Review-level reporting of adverse events by health condition category

Health condition category	AE	No AE	NR	Reported adverse events or side effects
Oral and dental conditions	2	14	37	Pain, discomfort, burning sensation, edema, erythema (13 RCTs); transient erythema or warmth at application site (< 5%).
Musculoskeletal disorders	3	9	10	Mild wrist pain and discomfort (1 RCT); increased disease activity and burning sensation in treated joints (2 RCTs); transient erythema (3–5%) and mild discomfort at application site (< 2%)
Oncology-related complications	0	3	5	-
Wound and lymphatic conditions	1	5	2	Tingling sensation (1 RCT)
Ocular conditions	2	1	1	Discomfort from brightness (1 RCT); excessive afterimage duration (1 RCT)
Dermatologic and hair conditions	2	0	1	Xerosis, pruritus, mild scalp irritation, warm sensation, acne (2 RCTs); paresthesia, mild urticaria, headache, dry skin, pruritus, scalp tenderness, irritation, warm sensation, eczema, acne (4 RCTs)
Respiratory conditions	0	0	1	-
Auditory conditions	1	0	0	Sudden deafness and dizziness (1 RCT)
Total	11	32	57	

AE adverse events, *NR* not reported, *RCT* randomized controlled trial

Counts represent systematic reviews rather than individual participants or adverse events. "NR" indicates that the review did not provide adverse-event information and should not be interpreted as the absence of adverse events. The reported events are presented as described in the included reviews

Table 4 AMSTAR 2 overall confidence summary (n = 100)

Overall confidence	n	%
High	9	9.0
Moderate	1	1.0
Low	30	30.0
Critically low	60	60.0

underlying evidence, and interpretation was further limited by the generally low methodological confidence of the included reviews.

Interpretation of key findings

Favorable review conclusions were most common for oncology-related complications, driven mainly by reviews of oral mucositis reporting reductions in its incidence, severity, or duration. This aligns with clinical practice guidelines that recommend LLLT for supportive oncology care [116, 117]. Oral and dental conditions contributed the largest body of review evidence, with favorable conclusions reported for post-gingivectomy healing, recurrent herpes labialis, and adjunctive periodontal care. These favorable conclusions are consistent with the broader literature suggesting the potential of LLLT to promote oral healing and reduce pain [118, 119]. However, conclusions varied across indications, indicating that a larger volume of evidence did not correspond to uniformly favorable conclusions.

Across all included reviews, treatment parameters varied widely (wavelengths, 400–1,100 nm; fluence, 0.0075–590 J/cm²). This heterogeneity may have contributed to inconsistent findings across conditions and underscores the need for standardized protocols for light-based therapies. Furthermore, the cumulative reported sample size should not be interpreted as the number of unique participants due to overlap among systematic reviews and the use of units other than individual participants in some dental reviews.

Adverse events and safety considerations

Adverse-event reporting was incomplete across the included reviews. Fifty-seven reviews did not provide adverse-event information, which should not be interpreted as indicating that no adverse events occurred. Reviews addressing harms generally lacked consistent denominators, comparator-group data, causality assessments, standardized severity classifications, and absolute risks. Although localized discomfort and skin reactions were reported, potentially important events, including sudden deafness with dizziness, increased disease activity, and prolonged afterimage duration, were also described. Consequently, no firm conclusions could be drawn about the incidence, comparative risk, causality, or overall safety of LLLT. Future trials and systematic reviews should systematically monitor and report adverse events, including comparator-specific denominators, severity, causality, and absolute risks, to enable more reliable safety assessments [120, 121].

Methodological quality and evidence certainty

Confidence in the review conclusions was limited because most systematic reviews received low or critically low overall confidence ratings using AMSTAR 2. Although many reviews performed well in core methodological domains, such as defining the review question, conducting literature searches, describing included studies, and assessing risk of bias, shortcomings in critical domains reduce confidence even when conclusions appear favorable. Limited protocol

registration increases the risk of selective reporting, the absence of excluded-study lists reduces transparency, incomplete reporting of funding sources limits assessment of potential conflicts of interest, and insufficient evaluation of publication bias may overestimate treatment effects. Because 90 of the 100 included reviews (90.0%) had low or critically low methodological confidence, the favorable conclusions should be interpreted cautiously and should not be viewed as evidence of high certainty.

Due to the substantial heterogeneity in outcomes and inconsistent reporting of pooled estimates across reviews, outcome-level quantitative synthesis and sensitivity analyses according to methodological quality were not feasible. Therefore, findings should be interpreted together with the overall methodological confidence. Although AMSTAR 2 provided an assessment of review conduct, it does not indicate the certainty of the underlying evidence or confidence in effect estimates.

Implications for clinical practice

Clinical application of LLLT should be considered on a condition-specific basis. Oral mucositis represents a particularly promising application, as favorable review conclusions were frequent and were consistent with existing clinical practice guidelines. Favorable conclusions were also reported for selected oral and dental, wound-related, musculoskeletal, ocular, and dermatologic indications, although findings were less consistent across these areas. Given the generally low methodological confidence of the included reviews and substantial variation in treatment parameters and outcomes, clinical decisions should consider condition-specific evidence and guidance, treatment parameters, and patient characteristics rather than the number of reviews reaching favorable conclusions. The available evidence did not permit firm conclusions about safety, and cost-effectiveness was not evaluated in this umbrella review.

Research gaps and future directions

Future research should address several key gaps. First, intervention parameters should be standardized to reduce heterogeneity and enable more reliable comparisons among studies. Wavelength, power, fluence, treatment duration, and frequency should be clearly reported in clinical trials. Second, larger, well-designed randomized controlled trials are needed. Third, future systematic reviews should improve methodological rigor through protocol registration, transparent reporting of excluded studies and funding sources, and assessment of publication bias. Finally, safety outcomes should be systematically assessed and reported, ensuring that risks are adequately evaluated alongside benefits.

Strengths and limitations

The primary strength of this umbrella review is its comprehensive synthesis of systematic reviews across diverse clinical applications of light therapy. By evaluating the evidence at the review level, this umbrella review provides a high–level overview of the breadth and credibility of the findings. However, this review has several limitations. Although the overall CCA was 0.48%, this aggregate may obscure overlap within particular clinical areas because it combines reviews addressing diverse conditions. Category-specific overlap was considerably higher for dermatologic and hair conditions, ocular conditions, and wound and lymphatic conditions. Pairwise analysis further identified substantial overlap between some individual reviews, with within-category pairwise CCA reaching 71.43%. Repeated inclusion of the same primary studies may therefore contribute to apparent consistency across review conclusions and should not be interpreted as independent confirmation of treatment effects. Although category-specific and pairwise analyses were performed, further stratification by specific condition, comparison, and outcome was not feasible because the available extracted data did not consistently link individual primary studies to these combinations. Establishing these links would have required additional extraction from the primary studies. Broad health condition categories may therefore still obscure overlap within particular clinical questions.

In addition, the certainty of evidence (e.g., using GRADE) was not evaluated at the umbrella level, which limits the formal assessment of confidence in the effect estimates. The descriptive classification of review authors' conclusions was intended to provide a broad overview across diverse clinical conditions but did not account for differences in sample size, methodological quality, or strength of evidence among reviews. Consequently, reviews with substantially different evidentiary weight contributed equally to the summary findings, potentially oversimplifying the evidence base. The findings should therefore be interpreted alongside the AMSTAR 2 and overlap assessments. Additionally, variability in outcome measures, patient populations, and intervention parameters limited the ability to draw firm conclusions for several conditions and also limited the assessment of dose-response relationships. Finally, publication bias could not be assessed at the umbrella level, leaving the possibility that favorable conclusions are overrepresented.

Conclusions

The included reviews reported favorable conclusions for several clinical conditions, with the clearest positive pattern observed for oncology-related complications, particularly

oral mucositis. These findings suggest that LLLT has potential clinical value in selected indications and identify areas in which further clinical development and research may be warranted. However, these descriptive patterns do not establish treatment efficacy or certainty of evidence. Interpretation remains limited by heterogeneity in treatment protocols, incomplete adverse-event reporting, the absence of an umbrella-level certainty assessment, and the generally low methodological confidence of the included reviews. Future research should build on these promising areas while prioritizing standardized protocols, rigorous study designs, and higher-quality evidence syntheses.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate This umbrella review does not involve the collection of primary data from human participants or animals. Therefore, ethical approval and informed consent were not required.

Competing interests The authors declare no competing interests.

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